SUSTAINABLE SHRIMP AQUACULTURE AND MANGROVE MANAGEMENT

A SURVEY COMMISSIONED BY THE UNITED NATIONS VOLUNTEERS PROGRAMME (UNV)

Prepared by

Arati Krishnamoorthy

and

Olivia De Jesus

(*Online Volunteers* for the United Nations Volunteers programme)

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TABLE OF CONTENTS

- Introduction……..……..…… 3
- Environmental, social and economic impact of shrimp farming……...……..… 4
- New approaches in intensive commercial shrimp aquaculture in existing pond areas to increase sustainability…… ..……11
- New approaches for designing intensive commercial shrimp aquaculture ponds to increase sustainability and make it environmentally friendly……...……..…14
- Examples and solutions in small scale sustainable shrimp aquaculture……....15
- Options for community-based mangrove rehabilitation……...……..……18
- Other potential income sources in a mangrove delta environment……..21
- Conclusion……..……...…..23
- Appendix I
- Appendix II
- Appendix III
- Appendix IV
- References

INTRODUCTION

Mangroves are the coastal equivalent of tropical forests on land. Mangrove forests serve as a natural habitat for a variety of animal species like fish and crustaceans. They are hence an important source of food and other natural resources like fuel and medicine for humans and have traditionally been the source of sustenance for the local populations.

Mangrove forests once covered 3/4 of the coastlines of tropical and sub-tropical countries. Today, less than 50% remain, and of this remaining forest, over 50% is degraded and not in good form. [8] There are many factors for this, including the charcoal and timber industries, urban growth pressures, and mounting pollution problems. Among the most significant reasons for the loss of mangrove forests in many countries over the last thirty years has been the establishment of industrial shrimp farms. The demand for shrimp, especially in the United States, Japan and Europe has risen in the last few decades, leading to the development of the lucrative intensive shrimp farming industry. This has been backed by local governments, the Asian Development Bank, World Bank and FAO, as a means for creating jobs, obtaining foreign exchange and alleviating poverty in the developing nations. However, in most cases, it has only led to increased unemployment. Although shrimp exports have made major contributions to the economy of the producer countries, the financial benefits are often reaped by a few large companies and fail to trickle down to the local communities. Due to a variety of reasons, intensive shrimp farming has only left the local communities more economically impoverished than before. Intensive shrimp farming has also had serious social and environmental impacts.

- Mangrove cover in Bangladesh's Chokoria Sundarban forest fell from 7,500 ha in 1976 to just 973 ha in 1988, largely due to shrimp farm development This deforestation has reportedly had socio-economic impacts on over 90% of the local community. [42]
- Estimates vary, but some place up to 50% - 65% of Thai mangrove losses due to shrimp farm conversion since 1975. The rate of mangrove loss in Thailand has been estimated to be as high as 6037 ha per year. [42]
- Around half of 279,000 ha of Philippine mangroves lost from 1951-1988 were developed into aquaculture ponds. [42]
- 269,000 ha of mangroves were reportedly converted to shrimp ponds in Indonesia between 1960 and 1990. Shrimp farming remains a major threat to Indonesia's mangroves and 50% of *Nypa* (mangrove palm) stands in Borneo's Mahakam River have been lost to aquaculture in the last 10 years. [42]

- Vietnam has reportedly lost more than 80% of its mangrove forests over the last 50 years, mostly due to intensive shrimp farming. [42]
- Nearly 50% of Ecuador's mangroves have been lost during the last 20 years; most of this loss has been attributed to shrimp farm development. [42]
- Shrimp farms are reported to have been responsible for the direct destruction of about one third of dense mangroves in the Gulf of Fonseca, Honduras. The current rate of mangrove deforestation in Honduras has been estimated to be as high as 2,000 - 4,000 ha per year, a rate that could be expected to lead to the total loss of all mangroves by 2020. [42]

This paper will deal with the environmental, social and economic impact of intensive shrimp farming. It will then explore cases where efforts have been made to increase sustainability in existing intensive pond areas. New methods developed to practice intensive shrimp farming, while being sustainable and environmentally friendly will be detailed. Examples and solutions from small scale shrimp aquaculture will then be presented. Finally, options for mangrove rehabilitation and alternate means of livelihood in a mangrove ecosystem will be given.

LESSONS LEARNT FROM INTENSIVE COMMERCIAL SHRIMP AQUACULTURE

In intensive shrimp farming, pond construction begins by cutting down the mangrove forests and digging diked ponds. Then, they are stocked with post larvae, at high stocking densities. In order to force the shrimp to feed continuously, the pond is lit all night. It is fed with formulated protein pellets and supplementary artificial feeds. To prevent from diseases, a number of chemical inputs as antibiotics, pesticides and detergents are also added. Due to the high stocking densities, a lot of toxic waste is generated in the ponds. Hence clean water is pumped in from the surrounding areas and the toxic water is released out, resulting in accumulation of wastes and degradation in the surrounding ecosystems. [3] The environmental, social and economic impacts of this system are dealt with in this section. [1] [3] [4] [9] [21]

ENVIRONMENTAL IMPACT

1. Loss of species due to loss of mangrove habitat

Mangrove forests are critical to local biodiversity, harboring plants and animals totally unique to mangrove ecosystems. Their destruction hence leads to loss of local species. Vital habitats have been permanently lost for fish, mollusks and crustaceans, as well as numerous birds, migratory and endangered species.

2. Endangering fish and other marine species

The destruction of the mangrove ecosystem leads to the loss of local fisheries due to the following reasons:

i. Shrimp post-larvae – used to stock ponds – are small and are harvested with fine nets, which also collect many other species. This unwanted 'by-catch' is discarded and usually dies. This process has the highest by-catch rate in the world, up to 20 lbs. of fish lost for every one lb. of shrimp larvae caught. Included in the by-catch are rare species like turtles and seahorses and lots of other fish. This contributes to declining wild fisheries, including decreases in wild stock of the very shrimp larvae required by the industry, which once thrived in the now vanished mangrove forests. [1]

ii. Farming carnivorous species like shrimp intensively or semi-intensively can require protein inputs of over double the weight of the farmed species produced. For shrimp, this feed comes primarily in the form of fishmeal and fish oil; meaning that additional pressure is placed on marine reserves and a valuable source of dietary protein is denied to coastal communities. In Thailand, nearly one million tonnes of fish are converted into animal feed annually, the majority being used to feed shrimp farmed for export. [1]

iii. Release of chemical and toxic waste from the shrimp farms in the neighboring waters causes pollution and changes the habitats of the fish in the local fisheries.

- For every fry of the tiger shrimp, *Penaeus monodon*, collected in India, an estimated 160 fry of fish and other shrimps are lost. At just three collection centers in West Bengal between 62 million and 2.6 billion by-catch fry are estimated to be removed from the ocean in the course of collecting shrimp fry each year.
- In Thailand, it is estimated that for every 1 kg of shrimp produced, 434 g of fish is lost due to habitat conversion alone. [1]
- Decreased catches of shrimp larvae have been associated with the conversion of mangroves to shrimp ponds in Ecuador. [1]
- Habitat conversion has been responsible for the mass death of crabs in Brazil. [1]
- Within 2-3 years of large-scale mangrove clearance in Kuala Muda (Kedah, Malaysia) fishermen reported a drop in income to one-sixth of its prior level. [1]

- A year after shrimp farms began operating locally, Indian fishermen in Ramachandrapuram reported that catches had declined to one-tenth of previous sizes. [1]
- Fishermen in Chokoria, Bangladesh reported 80 % declines in catches since mangrove destruction and the creation of dikes for shrimp farming. [1]
- In 1997, it was reported that Burma's mangrove had decreased by 271,000 ha since 1993, leading to a decline in coastal fisheries production of 190,000 tonnes annually.
- In Campeche state, Mexico, annual mangrove declines of 200 ha from 1980- 1990 caused yearly fisheries losses of US$140,000. [1]

3. Erosion of shorelines

The mangrove forests protect shorelines from erosion caused by hurricanes and floods and they capture sediments - thus protecting coral reefs. Destruction of the forests exposes the shorelines to the ravages of nature, causing coastal erosion, sedimentation, and water pollution, thereby affecting coral reefs, seagrass beds (and species like the green turtle and dugong that feed solely on these beds) and the productivity of coastal waters.

- In 1991, thousands died in Bangladesh when a tidal wave swept into an area whose mangroves had been converted into shrimp ponds. In 1960, the same area had been hit by a wave of comparable magnitude but intact mangrove forest dissipated its force and no lives were lost. [1]
- In 1999, when a 'super-cyclone' hit India, killing at least 10,000 people and making 7.5 million homeless, areas with intact mangrove forest were largely unaffected. [1]
- In 2001, a major storm hit Ha Tinh, Vietnam. In areas with mangroves, the coastal flood defense dikes were safe, whilst in those without mangroves dikes eroded away. In 1997, the biggest typhoon for 100 years hit the Mekong delta, with significantly less damage in mangrove areas. [1]

4. Pollution of neighboring waters

There is a vast volume of waste produced inside the ponds by the shrimps. Feed eaten by shrimps but not retained in their body ends up as waste. As the waste piles up, bacteria flourish and consume the available oxygen. This can suffocate the shrimps and reduce their growth. Intermediate waste products of both shrimp and microbes - such as ammonia and nitrite, are toxic to shrimp, fish and other animals. Shrimp weakened by waste and lack of oxygen is more susceptible to disease. In order to avoid this, water from neighboring rivers or groundwater supplies is pumped in and the wastewater of the

ponds is pumped out into rivers, canals or other near coastal waters. This system results in the pollution of the neighboring surface waters with this waste.

To guard against diseases farmers also use a large amount of antibiotics during production as well as toxic chemicals (pesticides, fungicides and algaecides) between harvests to sterilize the ponds. These chemicals also get into the neighboring waters during exchanges.

Since some species of shrimp require brackish water, the activity also leads to salts penetrating the water table eventually and provokes the salination of coastal aquifers and agricultural lands.
When the ponds are abandoned due to disease or other causes, the area is often left as a wasteland and the soils contain high levels of salinity, acidity and toxic chemicals, which make other uses practically impossible. Rehabilitation of these lands is then too costly for the local people and governments.

SOCIAL IMPACT

Displacement of local communities is common in shrimp exporting countries, where politically connected investors turn highly productive mangrove ecosystems on which a large local community is dependent, into a single use private domain. The many poor people who depend on mangrove and coastal fisheries for their livelihoods are eventually displaced. Conflict over land tenure rights is at the core of the conflicts related to shrimp farming. Supported by government agencies and police, companies force the local people to give up their land with inappropriate compensation or even with no compensation at all. Labor conditions on these farms are often bad – child labor is employed, women are subject to sexual harassment. [1] [2]

A case study on the social impact of intensive shrimp farming in Indonesia is given in Appendix I.

Impact on health

1. Depletion, salination and chemical pollution of water resources has led to serious problems accessing drinking water supplies.

- In Bangladesh, cattle dung is traditionally used as fuel. Poor quality drinking water and a lack of food in shrimp farming areas has increased incidences of cattle illnesses, such as diarrhea. Together with a rise in mortality, this has reduced availability of dung fuel, resulting in less frequent boiling of water, with associated increases in water-borne disease. [1]

2. Since a lot of antibiotics are used during production, human consumers of tropical shrimps produced in this way are eating food containing high levels of antibiotics. Some of the antibiotics that are used in shrimp farming are the ones used in humans, which might decrease the effectiveness of antibiotics against diseases.

3. Chemical inputs to the pond include known or potential carcinogens, mutagens and teratogens, which also find their way into the shrimp. Many of the substances used in this activity are prohibited in some countries due to their carcinogenic effects; but there are no regulations limiting the amount used in the producer countries.

4. Organophosphate pesticides used have high neurotoxicity, potentially affecting farm workers. Pesticides that find their way into human food can disrupt the human endocrine system.

5. Rotenone (*Derris* root) used to kill fish in shrimp ponds can induce respiratory paralysis.

6. Shrimp ponds, especially the abandoned ones, serve as a breeding ground for insects. In 2001, it was reported that 70% of the World Bank sponsored ponds were abandoned. [1]

- In Indonesia, in 1999, the spread of malaria in south Sumatra was partly attributed to the presence of many abandoned shrimp ponds, which became breeding grounds for Anopheles mosquitoes. [1]
- In Sri Lanka, villagers have reported increased prevalence of insect-borne disease in shrimp farming areas. [1]
- In Bangladesh, following conversion of mangroves to shrimp ponds near Sonadia, local people reported ‘unprecedented’ attacks by mosquitoes and other insects. [1]
- During the El Nino of 1999, malaria spread in South Sumatera partly due to abandoned shrimp ponds, which became an optimal environment for larva of Anopheles mosquito.

ECONOMIC IMPACT

1. Depletion of natural resources that sustain local livelihood

Destruction of the mangrove results in loss of natural resources that have varied uses for the local community. The most common uses of mangroves and their ecosystems are extraction of firewood, building materials, charcoal, tannin, drugs and honey.

2. Unemployment

Shrimp farming is not labor intensive. It has led to loss of employment in the agricultural sector. Studies conducted in the Philippines and in India show that much of the local employment generated by shrimp farms is temporary, requiring high labor inputs only to construct the ponds. After initial facility development is completed, the shrimp aquaculture industry is a capital-intensive, rather than labor-intensive industry.

- An intensive shrimp aquaculture business in India employs just 5 workers, while a rice paddy business of equal size requires 50 workers.
- Statistics from Ecuador are more stark – whilst a single hectare of mangrove forest provides food and livelihood for ten families, a shrimp farm of 110 ha employs just six people during preparation and a further five during the season.

3. Livelihoods of local fishermen lost

As mentioned earlier, intensive shrimp farming leads to the destruction of local fisheries. According to FAO figures, 50% of the world's fisheries were already depleted in the year 2001. [3] This has affected the local coastal economies, which are strongly dependent on fishing from mangroves.

- In Sri Lanka's Puttalam District, nearly 20000 lagoon fishers have moved to urban areas in search of work following the impact of shrimp farming on their traditional livelihoods.

4. Reduced agricultural productivity [1]

Due to the regular pumped water exchanges with the ponds, the surrounding areas have been contaminated with toxins and saltwater, affecting the soil composition and rendering land infertile

Excessive salination of the soil can affect the growth of vegetables and plants used for cattle fodder. This, along with salination and pollution of water resources, contributes to increased cattle mortality, which affects rural economies.

Degradation of agricultural land contributes to the further expansion of the shrimp farming industry. Poor landowners affected by salination often have little option but to sell their fields at deflated prices to aquaculture operators or to turn to shrimp aquaculture themselves.

- In several coastal provinces of Thailand, many of the shrimp farms were located close to paddy fields, which have been impacted by saltwater contamination.
- For this reason, shrimp farming areas of Bangladesh that previously produced food surpluses must now import rice from other regions - for example, in Satkhira, rice production declined from 40000 tonnes in 1976 to just 360 tonnes ten years later largely because of salt encroachment from shrimp pond canals crossing rice-fields

COST – BENEFIT RATIO

Shrimp farming was originally introduced as a means of reducing poverty, creating employment, and generating revenue. Indeed, global data suggest that 1.2-1.5 million full-time equivalent jobs have been created directly by shrimp aquaculture. Shrimp crops can be highly profitable – earning 30 times the profit of rice farming in Thailand, for example. However, there are serious externality costs associated with shrimp farming in terms of the environmental and social impact it has had. Economically, although it is more profitable than traditional agriculture and aquaculture practiced in the area, it appears that the economic benefits of shrimp farming do not always reach the local communities, but are reaped by the big companies that manage it. Also, not being labor intensive like traditional methods, and by destroying natural forest resources and fisheries, it has led to unemployment and destroyed the livelihoods of the local people.

Analysis of a mangrove system in Thailand revealed that conversion for aquaculture made sense in terms of short-term private benefits, but not once external costs were factored in. The substantial social benefits associated with the original mangrove cover – from timber, charcoal, non-timber forest products, offshore fisheries, and storm protection – fell to almost zero following conversion. The TEV [total economic value] of intact mangroves exceeded that of shrimp farming by about 70% (c. US$60,400 compared with US$16,700 per hectare). [Balmford, A. et al. 2002 “Economic reasons for conserving wild nature”. Science 297: 950-953]

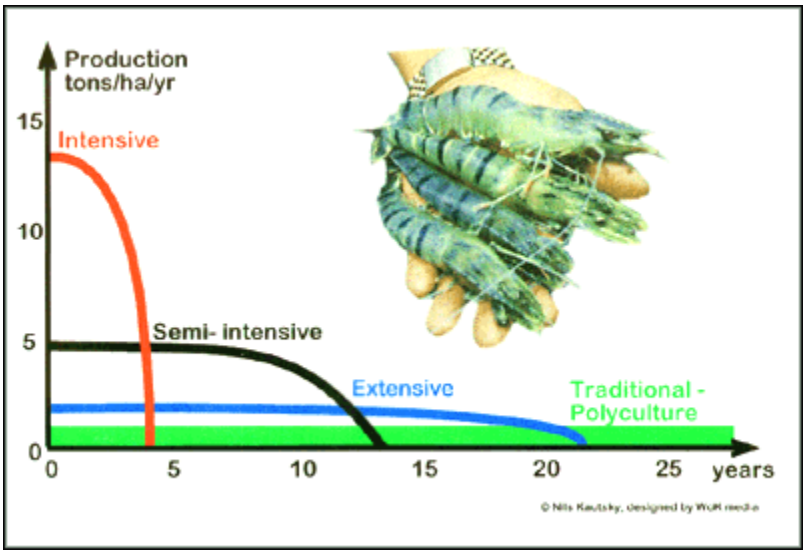
A study of the Matang mangrove in Malaysia revealed that its value for coastal protection alone exceeded the value of farmed shrimp by 170 percent. [3]

NEW APPROACHES IN INTENSIVE COMMERCIAL SHRIMP AQUACULTURE IN EXISTING POND AREAS TO INCREASE SUSTAINABILITY

As currently practiced, intensive and semi-intensive shrimp farming is often unsustainable.

- Data from India, Indonesia and Vietnam, for example, show that productivity rapidly declines and risks of disease outbreaks increase after 5-10 years of operation in these systems. In Sumatra, shrimp production declined from 10 tonnes / ha to 2-5 tonnes / ha over just four years. The result is that shrimp farms are often abandoned.
- In Thailand, it has been estimated that over 20% of shrimp farms located in former mangroves are abandoned after 2-4 years, and it has been estimated that as many as 50% of ponds in the country are disused.
- In 2001, it was reported that 70% of World Bank financed shrimp ponds in seven Indonesian provinces were abandoned.

The graph given below depicts the built-in unsustainability of intensive shrimp farming. [19]



Hence, companies currently operating in an intensive mode should consider taking measures to increase their sustainability. In this section we will discuss the measures to take to increase sustainability in existing pond areas, benefits to companies by taking these measures, and look at some relevant case studies.

MEASURES TO INCREASE SUSTAINABILITY & THEIR BENEFITS:

1. Sustainable generation of broodstock

In commercial ponds, broodstock is normally shrimp fry obtained from the wild and pond stocking densities are very high. Survival rates for such broodstock can be as low as 15% [35].
Broodstock needs to be generated domestically (as opposed to from the wild shrimp fry that is in many cases imported) and the pond stocking densities need to be reduced to semi-intensive or traditional extensive levels.

Generating domestic shrimp broodstock does away with the problem of high risk to diseases associated with wild or imported shrimp fry. Introduction of non-native species in the brood stock and the introduction of antibiotics in the ponds have led to outbreak of diseases and destruction of the very shrimp a few years later. For example, viruses introduced with non-native shrimp species have been linked to the 1993 crash of native shrimp crops in China. [1]. Lower stocking densities decreases production of waste, leading to increased pond productivity. [4]

2. Sustainable sources of food supply

The current practice of using fish and fishmeal as shrimp food is fast depleting their very source – the local fisheries. The high protein food also results in pond pollution. The feed conversion ratio can be as high as 3:1 [35]

The food used needs to have greater amounts of vegetable protein and less fishmeal. This makes the food more digestible, producing less waste in the ponds. Currently, the high waste generation makes the pond water toxic, prompting frequent water exchanges. The excessive organic matter (especially at night and early in the morning) is polluting as it blocks light and can inhibit algal oxygen production. At these times, dissolved oxygen content can be as low as 2.5 or 3 mg/l, whereas levels should be greater than 5mg/l in ponds2. It appears that a critical stage is reached around two months after stocking, when problems of disease often occur. By this time the farmer has invested heavily in feed, etc. and is in no position to attempt a remedy. [7]

Alternate sources of food with less fishmeal and protein content would also decrease the money spent by farmers on food. [9] [4]

3. Decrease and replacement of medication

The strong medicines and other chemicals used in the ponds to prevent diseases make the pond water highly toxic. Many of these chemicals cause stress to the shrimp, reducing their quality. Many of the chemicals find their way into the shrimp, and being carcinogenic are banned in the importing countries. The shrimp produced thus is unfit to be sold, or is sold at reduced prices.
Medication should be used only if necessary for the control of a specific disease for which the medication is thought to be effective. Shrimp farmers should follow information on product labels regarding dosage, withdrawal period, proper use, storage, disposal, and other constraints on the use of a chemical including environmental and human safety precautions. [35]

4. Better water management & nearby mangrove rehabilitation

The effluent from the ponds (which is high in toxic waste and chemical content) is discharged into neighboring water bodies, which in turn serve as a source of water for the ponds. It is hence in the interest of the shrimp farmer to filter the effluent. Often water exchanges with the neighboring water bodies introduces alien species in the tank and increases the risk of disease. Filtering and recirculating the pond water is a much better option, reducing the need for water exchanges. Water quality monitoring systems can also be installed as a check.

Biofiltration of the effluent can be done by the mangroves themselves. There are also numerous other options for biofiltration, including use of mollusks, seaweed, and finfish ponds. Recent studies have been done on the use of halophytes for treatment of saline aquaculture effluent, such as *Suaeda* and *Salicorna.* Halophytes are being used for effluent treatment in Eritrea. [39]

CASE STUDIES

The Colombia case study given in Appendix II involves discharging the pond effluent into a separate 1-km-long effluent canal, which then brings water into the mangrove wetland. The mangrove acts as a biofilter. An area of 120 ha of the mangrove is closed by levees, and water flow through the biofilter is controlled by concrete structures. Several exits distributed all around the biofilter direct water flow and recirculation, and allow the discharge of effluent into the neighboring Soledad Lagoon. An estimated 80%

of treated water is recirculated using this configuration. The main factors influencing the development of this effluent treatment system were the introduction of the blue-green algae into the tank due to the earlier practice of pumped water exchanges with the surrounding water, and effluent taxes imposed by authorities. Recirculation avoided the introduction of the blue-green algae into the tank, which was causing an off flavor in the shrimp, leading to financial losses. This system also led to the regeneration of mangrove trees in the biofilter area. [37]

The Amatikulu farm in South Africa is another example of a farm operating in semi-intensive to intensive conditions that experienced low survival rates at the start. The use of better stocking practices, feed management, water management has increased production significantly from the middle 1990s to 2000. [38]. More details on this case study are given in Appendix II.

NEW APPROACHES FOR DESIGNING INTENSIVE COMMERCIAL SHRIMP AQUACULTURE PONDS TO INCREASE SUSTAINABLITY

In the US, Thailand, and other countries where industrial shrimp aquaculture is being competitively pursued, a new alternative method is being lauded as more sustainable. This is the so-called "closed production system" approach. Studies show that shrimp farms that adopt closed water management systems tend to experience higher survival rates compared with open system farms. [18] This method involves recycling the production pond waters by removing toxins from the effluent waters. This is done either by complex and costly water filtration systems, or establishment of settlement ponds, or integrated secondary containment ponds. [20]

Integrated aquaculture techniques are also proving promising for semi-closed production methods. In some ponds, oysters and other shellfish, finfish, and seaweed are being cultivated either together with the shrimp, or in separate but interconnected ponds. The recycling shrimp pond water provides many nutrients for the other cultured species, which in turn can filter out a lot of the particulate matter and pollutants [20]

Two projects in this area, the ASEAN-SEAFDEC project [24] and the Belize Aquaculture system [36] are efforts in this direction. Detailed descriptions of these two projects are included in Appendix III

The principal benefits of these systems are: [20] [36] [24]

- The systems are much more efficient in land use that semi-intensive or extensive systems.
- Since the water is recirculated and often there is zero water exchange, the system is 20 to 40 times more efficient than traditional systems in terms of total water requirements.
- As no effluent is discharged to the surroundings, which serve as a source of water for the ponds, the system is not self-polluting.
- Due to water reuse and no discharge and as the bottoms are lined, the potential for pollution of coastal waters and salination of groundwater is greatly reduced.
- The system provides for the more efficient use of feed protein and fishmeal than conventional systems.
- Most of these systems may be constructed on any type of stable soil. This, coupled with low water pumping requirements, enable the ponds to be located in places outside of the immediate tidal zone. This prevents many of the negative impacts of locating traditional shrimp farms in coastal wetlands.

Due to the high setup costs involved, these high technology and capital intensive systems cannot replace in importance for developing nation coastal communities the more labor intensive and sustainable traditional systems. However, such closed-systems hold great potential to one day fill the current outside market demands of those numerous shrimp importing nations. [20]

EXAMPLES AND SOLUTIONS IN SMALL SCALE SUSTAINABLE SHRIMP AQUACULTURE

SILVOFISHERY [10]

Silvofisheries is a form of integrated mangrove tree culture with brackish water aquaculture. It is a form of low input sustainable aquaculture. This integrated approach to conservation and utilization of the mangrove resource allows for maintaining a relatively high level of integrity in the mangrove area, while capitalizing on the economic benefits of brackish water aquaculture.

Silvofisheries technology is labor-intensive, and quite appropriate for individual or family operations. An important factor is that this technology offers a viable alternative to other existing brackish water pond culture techniques, which are themselves too often unsustainable.

Two basic silvofishery models exist, the traditional Empang Parit Model and the Brackish water pond/mangrove alternation model.

The modern version of **Empang Parit Model** is today being promoted by the Indonesian Ministry of Forestry and the Directorate General of Fisheries (DGF). The mangroves are situated on a slightly raised central earthen platform within the aquaculture pond itself, which is bordered by raised dike walls and excavated shallow trenches, or canals, around 3-5 meters wide and 40-80 cm. below the central raised pond bottom. The ponds are designed so that water levels can fluctuate with the diurnal action of the tides, thus allowing periods when the mangrove roots are inundated with brackish water, then other periods when they are dry. However, within the pond's surrounding canals, which run adjacent to the dikes, pond water is retained full time, allowing for continuous extensive culture of fish, shrimp, and crabs within the pond facility. The density of the planted or naturally growing mangrove trees on the pond central platform may vary from 0.17 trees/sq. m. to 2.5 trees/sq. m., and properly adjusting this vegetation density actually contributes to the health and production capacity of the pond itself. The density influences the quantity of litter production and organic load in the pond along with other factors of cultivation. Less dense vegetation is suitable for culture of milkfish, while denser vegetation suits the culture of shrimp or mangrove crab.

The Empang Parit model represents the greatest level of reforestation or maintenance of existing forest (60-80%) to pond area. Though it offers a sustainable production model, this model poses some serious management challenges for the pond manager, including greater initial construction costs per unit of culture area, reduced penetration of sunlight onto the pond area due to the blockage from the mangrove canopy thus lowering the productivity of important phytoplankton and algae, limitation on species cultured, and reduced water circulation resulting in water stagnation and low oxygen levels.

The **Brackish water Pond/Mangrove Alternation Model** makes ponds more inexpensive to construct, offering a potential 400% (4-fold) greater income generating area for the same pond dike construction cost. It is also a suitable option for use in rehabilitating abandoned shrimp farms, thus saving on the expense of new pond construction, combining new pond construction with mangrove rehabilitation efforts involving the local community.

In this method, alternating areas of ponds would be interspersed with mangrove forest strips, which are aligned with a perpendicular orientation to the coastline, thus allowing nominal flow of freshwater to the mangroves within the area of the greenbelt. Typically,

for every two ha of pond area there would be an 8-ha mangroves maintained around the pond unit. (The ratio of mangrove to pond culture area is as high as any of the other methods, while providing superior management control and potential production). This model also allows for the maintenance of a greater natural biodiversity and a far more complex coastal wildlife refuge. The alternating brackish water pond/mangrove model is most appropriate for areas under private ownership or common control, such as government or community owned common areas.

TRADITIONAL SUSTAINABLE EXTENSIVE METHODS: [20]

There are some traditional methods of shrimp aquaculture, which have been practiced sustainably on a small scale for thousands of years, which are very similar to the silvofishery models described above. These systems are low-intensity, usually sustainable systems, which depend on diurnal tidal inundation to supply the larval shrimp and all of their food nutrients to the ponds. Excavation of shallow, relatively small ponds among the mangroves allows a containment area for juvenile shrimp to mature without much maintenance required. Since mangroves also serve as natural shrimp nurseries, there are sufficient supplies of shrimp larvae to naturally stock the ponds. Stocking rates in these traditional ponds are less than 10,000 fry per ha (<1 per sq. m.). Usually, these are polyculture ponds, also containing finfish, such as milkfish in combination with the shrimp. Yields are low, perhaps less than 500 kg per ha per annual harvest, but this provides additional supplemental income and protein source to make such production worthwhile. Traditional pond production mainly satisfies local consumer needs, as very little product is exported.

Indonesia's traditional "Tambak" System involves rearing of finfish and shrimp in the rice paddies after the rice has been harvested. The man-made dikes, which usually separate and protect the paddy from the incoming tides, are intentionally breached so that seawater can enter at high tide. In this way larval fish and shrimp are captured and reared to maturity. After the fish or shrimp are harvested, the paddies are reconverted in preparation for the next rice crop. This system of production may be one of the earliest forms of aquaculture practiced in Asia.

The Gei Wai System of Hong Kong is another traditional aquaculture system. Gei Wais basically utilize the positive attributes of natural mangrove forests as nursery and breeding grounds for fish, crabs, mollusks, and shrimps. Wide channels around one to three meters in depth are excavated around what becomes a small island of healthy mangrove forest.
The channels allow the several acres or more of each Gei Wai pond to hold sufficient waters at low tides to sustain the captured shrimp and fish. At high tides renewed

sources of nutrients enter the ponds through constructed sluice gates to sustain pond life anew. In this fashion, up to 1900 kg of shrimp can be raised and harvested annually from one Gei Wai. Today, there is only one remaining area of Hong Kong, called Mai Po, which borders Deep Bay, where gei wais are still found.

These traditional systems may not, in themselves, be viable at the scale of commercial shrimp aquaculture enterprises; however, they do away with the immense external costs of industrial shrimp aquaculture.

OPTIONS FOR COMMUNITY-BASED MANGROVE REHABILITATION

Protection of existing mangrove forests and restoration of those deforested are important elements of maintaining the integrity of the coastal environment. Several NGOs have been actively involved in promoting community based mangrove rehabilitation. Most of them have developed education programs and training courses for the local communities. Given below are the main organizations involved in mangrove rehabilitation and examples of training courses that are available. Detailed case studies are provided in Appendix IV. Common pitfalls observed during these efforts and ways to avoid them are also provided in this section.

Organization name	Region	Contact details
Mangrove Action Network (MAP)	Worldwide	P.O. Box 1854 Port Angeles, WA 98362-0279 USA Fax: 360-452-5866 e-mail: mangroveap@olympus.net web: http://www.earthisland.org/map/index.htm
Industrial Shrimp Action Network (ISA)	worldwide	14420 Duryea Lane, Tacoma, WA 98444 USA phone: (253) 539-5272 fax: 253- 539-5054 e-mail: isanet@shrimpaction.org web: http://www.shrimpaction.com/AboutISANet.html
Redmanglar (Mangrove Network)	South America	CODDEFFAGOLF Edificio Fiallos Soto, 2do. Piso, oficina 204, Tegucigalpa, Honduras Tel: 504-238-8216 Fax: 504-238-0415 Email: cgolf@redmanglar.org

		Web: www.redmanglar.org
Environmental Justice Foundation (EJF)	worldwide	5 St Peter's Street, London, UK, N1 8JD tel: +44 (0) 20 7359 0440 fax: +44 (0) 20 7359 7123 email: info@ejfoundation.org web: http://www.ejfoundation.org/index.html
Small Fishers Federation (SFF)	South Asia (particularly SriLanka)	Pambala, Kakkapalliya, Sri Lanka P.O.Box 01, Chilaw , Sri Lanka Tel: 032-48707 Fax: 032-47960 E-mail: sffl@sri.lanka.net
Vietnam Red Cross (VNRC)	Vietnam	82 Nguyen Du Str. – Hanoi Fax: 84-4 9424285 Email:vnrchq@netnam.org.vn Web: http://www.vnrc.org.vn/index_e.html
Oxfam Great Britain	worldwide	274 Banbury Road, Oxford, OX2 7DZ, UK Tel: 44 (0)1865 312610. Web: http://www.oxfam.org.uk

TRAINING COURSES: [40]

- In Vietnam, training courses are conducted regularly for farmers in Ca Mau, Lower Mekong Delta by the Division of Forestry, Department of Agriculture and Rural Development (DARD). The farmers are from both the State managed Forest and Fishery Enterprises (FFEs) and from private farms. The planting and maintenance techniques that they learn are put in to practice on their farms with good success.

- In Thailand, the community of Pled Nai Village in Trad Province have been trained in mangrove rehabilitation, maintenance and protection. These activities have been possible through the strong commitment of a local NGO, Yad Fon, which has helped to mobilize governmental and public support for this poor coastal community.

- In the central Philippines, the New Buswang, Kalibo, Mangrove Plantation is an example of a successful 70-ha reforestation project due, among other factors, to the

people's association KASAMA (Kalibo Save the Mangroves Association), a community-based organization initially comprising 27 families and a local NGO called USWAG. The local community leaders were trained in basic law and organization by the local NGO.

- In Senegal, there are training programs in the Saloum Biosphere Reserve on how to develop sustainable livelihoods, planting of mangroves, managing finances and organizing women's groups. These programs held by a local NGO WAAME focus on mangrove management and planting and oyster farming and fish smoking.

- The Aquaculture Department of the Southeast Asian Fisheries Development and Education Center (SEAFDEC) offers regional training courses in Coastal Resources Management

- APEC (Asia Pacific Economic Cooperation) Program is offering training programs on mangroves for teachers and students from 21 countries.

- The Center of Advanced Studies in Marine Biology, Annamalai University, India conducts a United Nations University International Course on Biodiversity in Mangrove Ecosystems. This is a two week course offered annually to train trainers, young professionals from Asian country Universities or other institutions that can replicate the work in their own country on their return after training.

- The Center for African Wetlands (CAW) based at the University of Ghana is a regional body for Africa that was commissioned in 2001 to offer training programs to African institutions on the sustainable management, legal issues, policy framework, networking and research of wetlands, including mangroves.

MEASURES TO AVOID COMMON PITFALLS DURING REPLANTING: [7]

Species selection and seedling density are of great importance and replanting efforts would benefit from:

- Increased information exchange between the different groups planting mangroves.

- Education of local communities about importance of mangrove areas and benefits they can bring, preferably prior to planting.

- Local communities' involvement in decision-making (e.g. species selection) and in the actual replanting process (which, if paid, can bring employment opportunities, if unpaid may simply promote a sense of ownership and pride). Farmers can be encouraged to grow seedlings for income generation, and this can also lead to increased likelihood of survival due to reduced transfer distances.

- Hiring community members to protect small mangrove areas.

- Incorporating local community rights to exploit forest products sustainably.

- Avoiding high density planting. (0.3-0.7m intervals) High density planting hinders root development and reduces trees' ability to withstand strong winds or tidal action. It also reduces sunlight and thereby reducing leaf decomposition rates, and reduces productivity and growth rates.

- Avoiding monocultures. Monoculture plantation encourages pest attacks thus reducing survivorship. Pesticides commonly uses are highly toxic to aquatic life.

- Avoiding planting on areas (such as mud-flats) which have significant populations of shorebirds or are ecologically important in their own right.

OTHER POTENTIAL INCOME SOURCES IN A MANGROVE DELTA ENVIRONMENT

Alternative livelihoods need to be identified and promoted. These will vary depending on locality but might include: [7] [10] [31]

- Lobster aquaculture

- Seaweed culture

- Bee-keeping

- Pearl culture

- Mangroves are used as an important and potentially sustainable source of **fuelwood and charcoal** to meet the increasing needs of developing countries for domestic

fuel. Interest has also arisen in species like *Nypa* palm, which produces alcohol that can be turned into transport fuel.

- **Rotating rice-shrimp production**.

Integrated fish and rice farming has been the backbone of traditional agriculture in Asia for centuries. This traditional system offers enormous potential for local food security and household nutrition. They also take advantage of the services that coastal ecosystems provide, such as filtering and purifying water, cycling nutrients, removing contaminants and buffering the land from coastal storms and severe weather.

- **Mudcrab farming using silvofishery [10]**

Shore-based pen culture of mangrove mudcrabs is a new aquaculture undertaking introduced by the Inland Fisheries Division, Department of Agriculture, in Sarawak, Malaysia to assist the artisanal fishermen increase their income. It is another variation on the silvofisheries model, whereby the vegetation is left intact to provide the natural environment for the mudcrabs to grow and reproduce. As such it is an ecologically friendly system.

Comparative analysis was made for various options, such as mangrove crab, seabass (Lates calcarifer), tilapia/chicken and milkfish/shrimp. In both those silvofisheries models described above, the production of mangrove crabs was found to be the most economically sound production. On a per unit production basis, the mangrove crab culture outperformed all other species by a substantial margin ($22.79 net profit per unit area for crab culture, compared to $0.13 for seabass, $0.26 for tilapia/chicken, $0.25 for milkfish and shrimp)

The dimensions of the pen are 18 m by 9 m covering an area of 162 sq. Inside the pen, perimeter drains 0.3 m wide and 0.3 m deep are constructed. These drains are linked to the inlet and outlet drains outside the pen. Today, each mudcrab pen is stocked with 1,000 to 1,500 crabs. This is a nominal range of stocking density, as higher stocking rates of up to 5,000 crabs had been earlier attempted, but had resulted in higher early moralities. The crabs are fed with trash fish. Feeding is done once a day during high tide. The average quantity of trash fish used per pen per production period is about 600 kg.

Water management and proper pen maintenance are important. Tidal flows and regular pen water exchange help maintain water quality, thus ensuring the health of the culture system. In 4 to 7 months, crabs reach harvestable size at around 300 g, and at least two harvests annually are possible. Restocking with small crabs (100 g) from the nearby river maintains the original number of crabs in the pen, thus optimizing a continuous

production. The average production per pen per production period is about 530 kg, and mortality rate about 47%.

Average net income per pen per month, at the average production period of 5.2 months, is about US$330 Participating fishermen have seen a marked increase in their income. Some have gone on to establish 4 to 8 pens, increasing their earnings potential by an additional US$1,320 to US$2,640. All this is possible while utilizing only a very small area of the surrounding mangrove forest, while at the same time helping to conserve this valuable natural resource.

In addition to helping to alleviate poverty in the fishing community, this system of crab culture offers two valuable bonuses:

i) Increased wild crab population - There is a strong possibility that the berried female crabs can hatch their eggs in the pen under the salinity condition during the high tide and release the larvae into the river thereby contributing to the recruitment of crabs into the Sematan mangrove ecosystem. This belief is reinforced by the fishermen noting an increase in the number of crabs in some of the tributaries of the Sematan River. It is believed that this recruitment is contributed by the pen culture system.

ii) Participating fishermen who had adopted the pond culture system were actually replanting mangroves in the degraded or bare areas in and around their pens to improve the mangrove canopy, which provided the desired shade and litter fall to help maintain the proper balance within the pen system itself, thus producing healthier crabs and reducing crab moralities. This has led to a healthier mangrove forest environment as well, which again increases both the area's biodiversity and the productiveness of the wild coastal fishery.

Note: Despite the encouraging preliminary results from this newly advanced mangrove crab culture system, the sustainability and impacts of this system should be thoroughly investigated.

CONCLUSION

The mangrove ecosystem is of great importance, both from an environmental stand point and from that of a means of livelihood of the local community. Past practices in intensive shrimp aquaculture in a mangrove environment have led to serious economic, social and environmental issues. Currently about half the original mangrove areas in the world have been destroyed, the primary reason being commercial shrimp farming.

In order to stop this destructive trend it is important first of all to take measures to increase sustainability in existing pond areas. This is done by treating the effluent waters before discharge and possibly recirculating them, having a sustainable generation of broodstock and reasonable stocking densities, moving from high protein fishmeal as food to more sustainable and low protein diets, and using lesser amounts of chemicals and medicines in the ponds. Several projects around the world have been investigating methods to achieve intensive shrimp aquaculture, while remaining sustainable and environmentally friendly. These mostly use a closed system with effluent waters being treated and recirculated. While not a viable option for most developing nations due to their high capital costs, they do hold a future potential to meet the demand in the shrimp importing countries themselves.

There are also several traditional shrimp culture models which are more sustainable and environmentally friendly. These include the silvofishery models and the traditional local models practiced in Indonesia and HongKong, which involve either having the mangroves in a raised mound within the pond itself or having alternating ponds and mangrove stretches. The mangroves serve as a biofilter, cleaning the effluent from the ponds.

Alternate livelihoods also exist within the mangrove ecosystem. One good example is mudcrab farming using silvofishery methods.

There are several organizations operating across the world today that are promoting the rehabilitation of destroyed mangrove forests. These efforts have been most successful when they have involved the local community. Education and training of the local communities have also gone a long way in the effort to preserve or restore the mangroves.

APPENDIX I

Social Impact of commercial shrimp farming in Indonesia

One of the unique characteristics of shrimp farming in Indonesia is the application of the 'Inti-Plasma' or NESS (Nucleus Estate Smallholders Scheme). About 70-80% of the shrimp export in Indonesia is owned by three large companies – the PT Central Pertiwi Bratasena (PT.CPB), PT Dipasena Citra Darmaja (PT.DCD) and PT Wahyuni Mandira (PT.WM). The company converts large tracts of wetlands into shrimp ponds and then sets up agreements with smallholders, who buy all the input for farming one or a few ponds from the company and then sell the harvest to the company. Theoretically, the smallholders are expected to pay back their debt to the company within 7-8 years and to become independent owners of the pond and a small home. In reality, all the conditions and prices are set by the company, the accounts are kept by the company and the smallholders get trapped into a vicious cycle of poverty and debt.

Wahyuni Mandira Co. now possesses 30,500 hectares and is planning to expand to 170,000 hectares. Prior to its operations in 1997, part of the land belonged to the local people and the other was a conservation area. 2,200 farmers were forced to give up their land for very small compensation, as the Provincial Government claimed that the land was a government asset and the local people didn't have land rights. Only 10% of them were invited to become smallholder farmers and the others were asked to migrate. More than one thousand resisted and stayed on in neighboring land and in mangrove areas.

Then, in November 1998, about 1,600 farmers (smallholders) protested against the conditions imposed by the company. Frustrated by lack of response by the company, the National Parliament in Jakarta, the Regional Government of South Sumatera, and the National Commission on Human Rights, the farmers started to demonstrate in front of the farm management office and the situation went out of control, degenerating into riot. Minutes after the riot started, the farm was surrounded by military, while the farmers were trapped inside the farm with no food for several days. At the end, more than 30 farmers were arrested, 16 of them sentenced to prison for periods from 6 months to 5 years. The court never considered that there was strong evidence that the riot was set up by the company.

In order to expand the farming operation to 170,000 hectares, during the year 2000 the company built water canals through local people's lands, creating further tensions and conflict. Some of the local people run traditional shrimp farming. The company moved in with the protection of the army and police.

Similar stories have been unfolding in Dipasena Farming, a nearby shrimp farm under the same holding company, where more than 1,700 farmers (smallholders) protested for the same reason and the same demand, and in Shrimp Banggai Sulawesi farm, a 100 hectare joint venture between an Indonesian and a French company. The local people have filed a case against the companies, but are still waiting for a response from the Lower Court. Other well-documented similar cases have taken place in Maluku, Papua, and other locations in Sumatera. Resistance from the communities is not only related to land rights, but also to environmental impacts. The community in Bengkulu, Sumatera, opposed the construction of a shrimp farm in their area because of environmental concerns.

On August 15, 2001, Central Pertiwi Bahari (CPB), also known as Bratasena Farming, in Lampung Provinces was recently charged by 147 local people for land rights conflict over 347 hectares of land. The local people claimed that their land had been occupied by the company in 1995 without any compensation. The local people complained about the case to the company, local government and National Parliament at that time, but there was no response. The company now insists that it will respond to the claim if asked to do so by the government.

The NESS system is also very biased against women. In large-scale shrimp farming only adult and educated men can hope to get a job. In case of death or inability to work of the smallholder males, women must leave the farming estate, leaving behind all the assets that they had been paying for by credit installment.

Impact on health and education can also be considered serious, particularly on children. During the El Nino of 1999, malaria spread in South Sumatera partly due to abandoned shrimp ponds, which became an optimal environment for larva of Anopheles mosquito. The lack of drinking water has caused a number of people in Wahyuni Mandira Farm, Sumatera, to suffer of pneumonia due to drinking rain water.

In conclusion, the change from traditional to industrial shrimp farming that is rapidly taking place in Indonesia might in the short term benefit the government and the large-scale shrimp investors due to foreign currency generation, but the environmental and social costs associated with the industry by far outstrip the benefits. Local communities are particularly marginalised and exploited in large-scale NESS farms and local social structures are threatened by growing tensions and conflicts.

APPENDIX II
NEW APPROACHES IN COMMERCIAL SHRIMP AQUACULTURE TO INCREASE SUSTAINABILITY IN EXISTING POND AREAS

A Case Study on the Caribbean Coast of Colombia – Integration of Mangrove and Shrimp Farming

Agrosoledad farm is located on the Caribbean coast of Colombia, about 10 km inside an estuarine area influenced by the Sinu River. Shrimp production first started in 1987. Pumped water exchanges were done with the waters in the Soledad Lagoon.

In 1994, Agrosoledad Farm experienced, for the first time, a problem of off-flavor in its raised shrimp, resulting in financial loss (harvest had to be delayed or sale price of shrimp was lowered). The off-flavor was perceived as earthy or musty, and the cause is thought to be the presence of some blue-green algae or actinomycetes in pond waters. Those algae were also present in Soledad Lagoon. The phenomenon appeared during the rainy season, when salinity dropped to very low values.

Blue-green algae usually flourish in tropical brackish water environments when salinity is low and particularly in case of eutrophication. Shrimp farm effluent could also contribute to the enrichment of the lagoon in nutrients. It also posed a risk of self-pollution by introducing blue-green algae into shrimp ponds via the pumped intake waters.

The best remedy to the situation was then to recirculate waters inside the farm as soon as salinity dropped below 10 ppt. Recirculation provided several advantages: avoiding introducing blue-green algae from Soledad Lagoon into the farm, preventing the discharge of blue-green algae into the lagoon in case of bloom at the farm, and avoiding contributing to the eutrophication of the lagoon. But pond effluent had to be treated to avoid self-pollution as defined above.

Then Agrosoledad's staff proposed the idea of using a 120-ha mangrove forest, included in the property, as a biofilter to treat pond effluents. Based on the principles of wetland treatment, this mangrove filter was expected to remove algae and nutrients by sedimentation and absorption-consumption, respectively.

Pond discharge waters are collected into a 1-km-long canal that brings the effluent into the biofilter. An area of 120 ha has been closed by levees, and water flow through the biofilter is controlled by concrete structures. Several exits distributed all around the

biofilter direct water flow and recirculation, and allow the discharge of effluent into Soledad Lagoon.

In the recirculation configuration, northeastern and southwestern exits are closed, as is the entrance of the supply canal. Water goes out through the northern exits into the recirculating canal and then the supply canal. An estimated 80% of treated water is recirculated using this configuration.

When farm staff wants to discharge treated waters into Soledad Lagoon and recirculate only a small proportion of it, the supply canal is kept open as well as the northeastern and southwestern exits. In this case, treated waters flow through additional mangrove–an estimated 260 ha–before going back into Soledad Lagoon.

Two other benefits were observed:
Abundant fauna was observed in Agrosoledad's biofilter, including primates, birds, reptiles, fish, crabs, and insects–a typical mangrove-associated fauna ecosystem Curiously, a very large bird community (thousands of birds) settled in the middle of the biofilter, in the 11-ha Conocarpus erectus area. The main species are egrets, herons, and neotropic cormorant, white ibis, and anhinga; even wood stork appears from time to time.

The other positive impact of shrimp pond effluents has been observed on growth and regeneration of mangrove trees in Agrosoledad's biofilter after four years. Initial measurements of Agrosoledad Farm's effluent performed by both the Autonomous Regional Corporation and the independent laboratory Cartagena Waterworks indicated that BOD (Biological Oxygen Demand) and TSS (Total Suspended Solids) values were, respectively, 70% and 51% lower in the biofilter effluent (sampled at the northeastern exits) than in supply water.

Shrimp Aquaculture in South Africa

There are only 2 commercial prawn farms in Kwazulu Natal. Both farms are managed by Amatikulu Prawns (Pty) Ltd. Both farms operate at semi-intensive to intensive conditions. Ponds are stocked once a year (one farming cycle per annum). One of the farms operates 10 ha of ponds, the other 26 ha. Both farms experienced low survival rates and have adopted better practices in order to improve the situation.

Current production practices include aquamats. In addition, improvements in early broodstock nutrition have been introduced thanks largely to discussions with people on the online Shrimp Discussion Group. Stocking rates are 25 – 40 PLs/m² with a grow-out

time of 6 to 8 months. Average yields have been about 3 MT/ha and 7 MT/ha, respectively, at each of the farms.

Shrimps are fed 4 to 6 times a day with constant monitoring. In the past, over feeding by new, untrained staff resulted in feed conversion ratios that were in excess of 5:1 in most ponds. While this problem has been addressed, it leads to the need to increase water exchange. Feeds are currently produced at Amatikulu farm and FCR yields are poor, ranging from 2-3 kg of food for every 1 kg of shrimp harvested. Diets produced locally have been constantly adjusted in terms of raw material composition, its source of supply and micro-nutrients, in an attempt to improve the FCR at the farms.

In general, pond water quality is managed with both aeration and water exchange. Due to organic loading, organic waste and sediments accumulated on the bottom of ponds are periodically sucked from the small, lined ponds and disposed on land. The unlined ponds are cleaned at the time of harvest. Eliminating the organic matter is much easier in the lined ponds. Since there is only one production cycle per year, down time after the harvest is not an issue.

Production in the last 5 years has increased substantially to just below 130 MT in 2000. Some fluctuation in total production during mid 1990s resulted from changes in management and shifts from imported Taiwanese feed to locally produced feed. Initially, at least, this feed was of low quality. Both farms may well increase their yields, but annual production is not likely to exceed 200 MT over the next years.

APPENDIX III
NEW METHODS DEVISED FOR INTENSIVE SHRIMP AQUACULTURE IN ORDER TO MAKE THEM SUSTAINABLE AND ENVIRONMENTAL FRIENDLY

ASEAN-SEAFDEC Project [24]

The Philippine-based SEAFDEC Aquaculture Department (SEAFDEC/AQD) and ASEAN have started a project to develop sustainable shrimp culture. The shrimp project is funded by the Japanese Trust Fund. Collaborators include Thailand's Department of Fisheries and Vietnam's Research Institute for Marine Products. [24].

Project sites are the Philippines (semi-intensive and intensive), Thailand (intensive), and Vietnam (semi-intensive). The activity of the Dumangas, Iloilo in the Philippines deals with the recent approaches to sustainable shrimp farming. The specific studies include partial discharge in intensive shrimp farming, zero discharge intensive shrimp farming, and intensive close recirculating ponds, and use of mangroves as filters for shrimp/fish pond effluents. Four activities are being implemented in Thailand. Seawater irrigation for intensive marine shrimp farming at Kung Krabaen Bay, Chantaburi; mangrove plantation for enhancing natural web in water recycle shrimp farm at the Chacheangsao Coastal Aquaculture Development Center, Chacheangsao; integrated physical and biological technologies for water recycle in shrimp farming at the Marine Shrimp Research and Development Center, Songkla; and mitigation measures of effluent from shrimp farming in mangrove and coastal at the Andaman Marine Shrimp Research and Development Center, Phuket. The activity in Vietnam is on the semi-intensive mangrove-friendly shrimp culture, which is implemented at Phu Long, Cat Ba Island, Cat Hai District. [25]

To further refine environment-friendly shrimp culture techniques that are known to be successful, two research activities have been approved for implementation under the revised mangrove-friendly culture project. The first is on nutrient cycles in order to provide the basis for understanding the complex inter-relationships of the bio-physico-chemical processes occurring in the pond. Initial findings have attributed the declining productivity in shrimp ponds to the deteriorating quality of pond inlet waters and the indiscriminate dumping of organic loads in inlet and drainage canals that go back to the body of water which is used as pond water for the shrimp culture, and inefficient water quality management in ponds. The second is on the capacity of mangroves to absorb nutrients and act as a biofilter. [27]

Belize Aquaculture Ltd: A Superintensive Shrimp Aquaculture System [36]

Belize Aquaculture, Ltd., has developed a superintensive shrimp aquaculture system. The main features of this system are:

- Low protein feeds: Low-protein feed creates a high abundance of organic particles for bacterial growth and floc formation. Turbidity caused by the floc eliminates light and prevents the growth of phytoplankton after the initial weeks of culture.
- Heavy mechanical aeration: aeration maintains high concentrations of dissolved oxygen in the pond water. High concentrations of dissolved oxygen promote aerobic decomposition of organic matter by bacteria, allow efficient and rapid conversion of ammonia nitrogen to nitrate through bacteria nitrification, and prevent low early-morning dissolved oxygen concentrations typical of phytoplankton-based culture methods; such low concentrations stress the shrimp
- Lined ponds: aeration is possible because erosion of the mineral-rich soil on the bottom is prevented by the plastic liner.
- Water recirculation and zero water exchange
- Sludge removal: All sludge removed is organic, because the plastic liner prevents bottom-soil erosion. Sludge is removed to prevent the accumulation of sludge in the centers of ponds; such accumulation would result in zones of anaerobic decomposition common in traditional aquaculture ponds with earthen bottoms.

The benefits of this system are:

- The environmental benefits of the Belize system are considerable compared to traditional semi-intensive systems. Because of water reuse the system is 20 to 40 times more efficient in total water requirements. Since it is not necessary to discharge effluents into the coastal waters which serve as a source of water for the farms, the system is not self polluting unlike many other types of shrimp culture systems.

- The Belize system is over five times more efficient in land use than semi-intensive systems. These two factors alone could substantially reduce the environmental impacts of shrimp aquaculture if this innovative system were adopted more widely.

- This system also provides more efficient use of feed protein and fishmeal than conventional systems.

- In the Belize Aquaculture system, grass could be established on embankments to reduce erosion. Because the ponds are well above the tidal zone, the soils are less saline. In conventional shrimp aquaculture systems, embankments are difficult or impossible to plant with short grasses due to the high salinity of the soils. Erosion of embankments is a major source of suspended solids in the effluent from conventional shrimp farms.

- In addition, the Belize Aquaculture system may be constructed on any type of stable soil, because the liner prevents seepage and water contact with the soil. Soil limitations (sandy, acidic, or organic soils) can be major problems in conventional shrimp ponds. Compatibility with a wide range of soils and the low water pumping requirements allow the system to be sited on land outside of the immediate tidal zone. This advantage prevents many of the negative impacts of traditional shrimp farming that result from locating shrimp farms in coastal wetlands.

APPENDIX IV
OPTIONS FOR COMMUNITY BASED MANGROVE REHABILITATION – CASE STUDIES

Vietnam Red Cross (VNRC) - Vietnam [7]

In implementing its successful planting schemes in Vietnam, the VNRC mobilizes locals, selecting the most vulnerable, and pays them small amounts of cash for planting. In addition, there is an educational element on how to plant, how to protect and why. The annual budget for the scheme is US$700,000 (provided by Danish and Japanese Red Cross Societies). To date 45,000 rural poor have taken part in replanting. Since 1994, 16,000 ha have been planted in 24 districts of eight provinces. The replanting has been highly successful, and has led to an increase in biodiversity in planted areas – especially with respect to zoobenthos, marine species and birds. VNRC's approach is to pay protection fees to local guards for three years after planting. These guards hail from local communities and deter clearance of the seedlings or damage due to cattle grazing. The guards are equipped with loud speakers, coats, and sirens and the regulations are posted on a noticeboards on the dykes. If people (or their water buffaloes) violate rules, they can be stopped and must face the risk of fines from the local People's Committee.

Oxfam Great Britain - Vietnam [7]

In Dong Hai Commune, Tra Vinh Province of Vietnam, Oxfam GB began mangrove replanting and promotion of integrated shrimp-mangrove farming in 1997. Mangrove area on the 2,230 ha site has since increased from approximately 2% to 60%15,16. The project has been very successful – survival rates of replanted trees are 82-97%16, soil and water quality in the newly reforested area has improved, and local farmers estimate that crop failure rate has fallen from 60% prior to 1997 to 20% today. Farmers have been involved in every stage of the project cycle, from designing to implementing, and the project has both raised awareness about the ecological roles of mangroves and created solidarity among the farmers . At the beginning of 2002, Oxfam began a more intensive education program and group organizing of farmers. Crucial in this project is the involvement of farmers in every step, including choosing species to replant. Another important lesson has been that farmers should grow seedlings themselves if possible, allowing them to generate income and reducing transfer distance for the plants.

Yadfon's Way - Thailand [11]

In Thailand, a small NGO in Southwestern Thailand, Yadfon (Rain Drop) Association, located in Trang, has taken up the cause for the mangroves and the coastal communities which depend on healthy mangrove ecosystems for their lives and livelihoods. For over a decade, Pisit Chansnoh, a cofounder and current President of Yad Fon, has led his organization in pioneering grassroots methods enhancing local community involvement in management of coastal resources. And, in those years of practical fieldwork, Khun Pisit and his dedicated staff have developed a methodology of village-level organizing that seems to be working.

Aku Cinta Indonesia(ACT) – Indonesia. [29]

This case study of community-based forest recovery shows that local institutional initiatives are very effective compared with programs developed elsewhere by the Forestry Department of Indonesia. The mangrove forest rehabilitation project in East Sinjai was based on a local group named the ACT for Aku Cinta Indonesia, or I love Indonesia. The government of Indonesia, through regional forestry officials, had attempted some conservation and rehabilitation efforts, but with success as low as 15 to 40%, due to factors like less control after replanting, poor quality of seed, site unsuitable for species planted and lack of community involvement. The first three factors were easily solvable. However, the last factor was outside the control of the rehabilitation program implementor. Realizing the need for a community based effort, the ACT started some groups to achieve this.

The most crucial factor, they realized, was having strong group leadership by a local person. From an informal local leader, most of the community is self-motivated to do conservation. However, only a few are motivated by the government instructions of most projects in Indonesia.

Specific doctrines ((1) need for coastal protection, (2) sustainable products for firewood and construction, (3) income generation, and (4) creation of new land) were adopted to serve as the theme and hope for the community.

The main program of these local groups was planting mangrove seedlings from September to December, when wave action is minimal. The seed supply came from local, ten year old trees. The seedlings are mature when they are 45 to 70 cm long. They were kept under the houses and covered with fabric, to reduce the smell and the attraction of crabs that eat them. The seedlings were planted 50 cm apart.

The group coordinated rehabilitation with other community institutions in the area. The end result was the development of mangrove forests for ecotourism and agroforesty,

using the empang parit or drainage ditch pattern, and increasing the income and welfare of the poor.

The Mangrove Action Project (MAP) - Caymen Islands [32]

MAP's efforts are unique in that they place the local community at center-stage in the restoration and management process. They believe that this "bottom up" approach makes a more effective and lasting difference than the conventional "top down" approach.
MAP will soon release its Mangrove Educational Curriculum, which was developed in the Cayman Islands for school children from kindergarten to ninth grade. They plan to introduce this curriculum to other parts of the world, modifying it to fit the local regions and translating it into the local languages. This will increase the appreciation of the mangrove ecosystem among local youth, who will later have to decide how to conserve and sustainably manage their coastal forests.

The Small Fishers Federation (SSF) – Sri Lanka [32]

The Small Fishers Federation (SFF), which is based in Pambala, near Chillaw Lagoon, carries out a range of activities to conserve Sri Lanka's mangrove ecosystem. SFF's Mangrove Resource Center at Pambala has become an important and respected education and research center for the country. The center's education programs have attracted school children, villagers, fisherfolk, teachers, local politicians, and government officers. The center now is conducting pilot projects in eco-friendly shrimp farming, bee keeping and crab-rearing in mangrove habitats. Mangrove replanting and rehabilitation efforts also have proven successful.

In The Hands Of The Fishers(IHOF) Program [32]

Last year, working closely with SFF and Thailand's Yadfon, MAP helped launch a program called "In The Hands Of The Fishers" (IHOF) - a series of workshops for grassroots NGOs and fisherfolk from three or more developing nations that contain mangroves. These workshops offer a format for information- and skill-sharing, while offering a toolkit of working alternatives to help enhance community based integrated coastal resource management. Follow-up projects at participating villages provide models of sustainable, low-intensity development alternatives. Past workshops have drawn participants from Thailand, Sri Lanka, Cambodia, Indonesia, Malaysia, India, Bangladesh, Pakistan, and Nigeria.

To demonstrate its partnership with MAP, the SFFL has renamed its Pambala facility the MAP-SFFL Resource Center. The fifth IHOF seminars were held at the center in April 2000. A second joint Mangrove Resource Center was inaugurated at Kiralakele in southern Sri Lanka during the April workshops, which involved representatives from Sri Lanka and a dozen West African countries, including Cameroon, Senegal, and Cote d'Ivoire.

IHOF workshops include field trips to nearby local fishing communities where delegates can see alternative development projects firsthand and talk to local villagers about their successes and failures in developing these small-scale programs. During these IHOF workshops, common strategies and new working alliances are developed to help coastal communities protect their mangrove forests, sea grasses, coral reefs, and near-shore fisheries.

Mangrove rehabilitation in Senegal

In Senegal, a program promoted by the National Parks integrates scientists, NGOs, women, youth camps and technical services in the management and rehabilitation of mangroves so that they can exchange experiences. The NGO WAAME is presently working with 50 villages, women’s groups and schools in reforesting mangroves. WAAME is also providing exchanges of experiences in fish smoking, oyster farming through community exchanges and local technical training.

National coordination committee - Ecuador

In Ecuador, there is communication and links between the different communities and user groups for example between North and South Esmeraldas, and between other provinces, and with the National coordination committee for mangrove protection. This system has been operating since about 1992 with effective results in promoting sustainable mangrove management in Ecuador.

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